

CYBER SECURITY INTERNSHIP PROJECT REPORT

Student Cyber Security Management System

Submitted By:

Name: Priya Kumari

Internship Program: Cyber Security Internship

Organization: White Band Associates

Project Title: Student Cyber Security Management
System

Technology Used: Python Programming Language

Abstract

The Student Cyber Security Management System is a Python-based application developed during the Cyber Security Internship program. The project integrates programming fundamentals with basic cybersecurity concepts to create a practical management system for handling student records and evaluating security practices.

The system allows users to manage student information, perform security assessments, analyze security scores, generate reports, validate login credentials, evaluate password strength, and maintain records using file handling techniques. The project demonstrates the application of programming concepts such as variables, conditional statements, loops, functions, lists, file handling, searching algorithms, and data processing.

Introduction

Cybersecurity has become an essential component of modern computing systems. Organizations require efficient methods for managing user data, monitoring security practices, and assessing potential risks.

This project was developed as the final internship project to combine all concepts learned throughout the internship. The system provides a simple but effective platform for maintaining student information while incorporating cybersecurity-related utilities such as:

- Password Strength Checking
- Login Validation
- Security Assessment
- Security Report Generation
- Student Record Management

The project demonstrates how programming can be used to automate administrative and security-related tasks.

Objectives

The primary objectives of the project are:

1. To develop a menu-driven Student Management System.
2. To implement file handling for permanent data storage.
3. To perform security assessments using predefined criteria.
4. To generate reports based on student security data.
5. To apply cybersecurity concepts in practical programming.
6. To strengthen problem-solving and logical thinking skills.
7. To demonstrate concepts learned during the internship.

Technologies Used

Programming Language

- Python 3.x

Development Tools

- Visual Studio Code
- Git
- GitHub

Concepts Implemented

- Variables and Data Types
- Input and Output Operations
- Conditional Statements
- Loops
- Functions
- Lists
- Searching Techniques
- File Handling
- Cyber Security Fundamentals

Project Overview

The Student Cyber Security Management System is a menu-driven application that performs multiple operations related to student data and cybersecurity management.

The system stores student information and evaluates security practices through a scoring mechanism. Users can add, search, view, and delete records while generating security reports.

System Modules

Module 1: Add Student

This module allows users to add student records.

Information Stored

- Student Name
- Student ID
- Branch
- Email Address

Purpose

To maintain a database of students for further security assessment and reporting.

Module 2: View Students

This module displays all student records stored in the system.

Features

- Reads data from storage
- Displays records in a formatted structure
- Improves accessibility of stored information

Module 3: Search Student

Users can search records using:

- Student Name
- Student ID

Benefits

- Quick retrieval of information
- Efficient data management

Module 4: Delete Student

This module allows deletion of records.

Features

- Search existing record
- Delete selected student
- Confirmation message displayed

Module 5: Security Assessment

The Security Assessment module evaluates the security awareness of a student.

Parameters Evaluated

- Multi-Factor Authentication (MFA)
- Password Length
- System Updates
- Antivirus Installation

Security Score Generation

Each parameter contributes to the overall security score.

Categories

Score Range	Security Level
90–100	Excellent
75–89	Good
60–74	Moderate
Below 60	Poor

Module 6: Report Generation

The reporting module generates overall statistics.

Information Included

- Total Students
- Individual Security Scores
- Average Security Score
- Students with Poor Ratings

Benefits

- Security monitoring
- Performance tracking
- Risk identification

Module 7: File Handling

The project uses text files to store information.

Operations

- Save Records
- Read Records
- Update Records
- Delete Records

Advantages

- Permanent storage
- Data persistence
- Simple implementation

Cyber Security Features Implemented

Password Strength Checker

The system evaluates passwords using:

- Length Validation
- Uppercase Character Detection
- Numeric Character Detection

Security Levels

- Weak
- Moderate
- Strong

Login Validation

A basic authentication mechanism verifies:

- Username
- Password

Benefits

- Access control
- User authentication
- Security awareness

Username Generator

Automatically generates username suggestions based on:

- First Name
- Last Name

- Birth Year

Example

- priya2024
- p.kumar24
- kumar_priya

Internship Tasks Covered

Task 01: System Information & Basic Automation

Implemented:

- Environment Setup and Python Installation
- System Information Display Program
- User Input and Summary Program
- Basic Calculator Application
- System Information Retrieval Utility

Features Developed:

- Displayed personal and internship details
- Accepted user input and generated formatted output
- Performed arithmetic operations (Addition, Subtraction, Multiplication, Division)
- Retrieved operating system information
- Displayed current username and working directory
- Displayed installed Python version using system modules

Skills Learned:

- Python Fundamentals
- Variables and Data Types
- Input and Output Operations
- Arithmetic Operators

- Python Modules (os, platform, sys)
- Basic Automation Concepts
- Program Structure and Execution

Learning Outcome:

Task 01 provided a strong foundation in Python programming by introducing core concepts such as variables, user interaction, arithmetic calculations, and system information retrieval. It also demonstrated how Python can automate routine tasks, which is a fundamental skill in cybersecurity and system administration.

Task 02: Decision Making and Logic Building

Implemented:

- Even/Odd Checker
- Largest Number Finder
- Grade Calculator
- Login Validation
- Password Strength Checker

Skills Learned:

- Conditional Statements
- Logical Operators
- Input Validation

Task 03: Loops and Automation

Implemented:

- Multiplication Tables
- Number Analysis
- Pattern Printing
- Password Attempt Simulator
- Username Generator

Skills Learned:

- For Loops
- While Loops
- Iteration
- Authentication Logic

Task 04: Functions and File Handling

Implemented:

- Calculator Functions
- Student Information Manager
- Marks Analysis System
- File Handling Operations
- Password Vault Simulator

Skills Learned:

- Functions
- Modular Programming
- Data Storage

Task 05: Arrays and Data Processing

Implemented:

- Marks Analyzer
- Employee Search
- Number Statistics Tool
- Login Log Analyzer
- Blacklisted IP Checker

Skills Learned:

- Lists

- Searching
- Data Analysis
- Security Monitoring

Task 06:

Implemented:

- Student Cyber Security Management System
- Security Assessment Module
- Report Generation Module
- File Storage System

Skills Learned:

- Project Development
- Integration of Concepts
- Cyber Security Fundamentals

Task 07: Employee Management System (Object-Oriented Programming)

Implemented:

- Employee class with ID, Name, Department, Designation, and Salary attributes.
- Add, view, search, update, and delete employee records.
- Salary statistics including highest, lowest, average salary, and total employees.
- Export employee data to CSV/TXT and read the stored records.
- Menu-driven interface with input validation and exception handling.

Skills Learned

- Object-Oriented Programming (OOP)
- Classes and Objects
- Constructors (`__init__`)

- Methods and Encapsulation
- CSV/File Handling
- Data Management

Task 08: Smart File Organizer

Implemented:

- Directory selection with path validation.
- Automatic file scanning and categorization by extension.
- Creation of Images, Documents, Videos, Audio, Archives, Programs, and Others folders.
- File statistics and categorized summaries.
- Search by filename and extension.
- Duplicate file detection.
- Automatic generation of file_report.txt.
- Robust exception handling for invalid paths, permissions, and file conflicts.

Technologies and Modules Used

- Python 3
- os
- shutil
- collections
- datetime
- Exception Handling
- File Handling
- Automation

Learning Outcome

This project provided practical experience with Python automation, directory management, reporting, and real-world file organization, improving problem-solving and software engineering skills.

Program Flow

1. Start Application

2. Display Main Menu
3. Select Desired Operation
4. Perform Data Processing
5. Store or Retrieve Data
6. Generate Security Assessment
7. Display Results
8. Return to Main Menu
9. Exit Program

Challenges Faced

During development, several challenges were encountered:

File Management

Managing student records efficiently while ensuring proper updates and deletions.

Security Score Calculation

Creating a fair and balanced scoring system for evaluating security practices.

Data Validation

Handling incorrect user inputs and preventing program crashes.

Code Organization

Maintaining modular and readable code using functions.

Learning Outcomes

Through this internship project, the following skills were developed:

Programming Skills

- Python Fundamentals
- Structured Programming
- Modular Design

Problem Solving

- Logical Thinking
- Algorithm Development
- Data Processing

Cyber Security Knowledge

- Authentication Concepts
- Password Security
- Security Monitoring
- Security Awareness

Software Development

- GitHub Repository Management
- Documentation Writing
- Project Organization

Future Enhancements

The project can be enhanced by implementing:

- Database Integration (MySQL)
- Password Encryption
- User Roles and Permissions
- Graphical User Interface (GUI)
- Email Notifications
- Advanced Security Analytics
- Cloud Storage Integration

Conclusion

The Student Cyber Security Management System successfully combines programming fundamentals with cybersecurity concepts learned throughout the internship. The project demonstrates the practical implementation of data management, security assessment, authentication, file handling, and report generation techniques.

This project strengthened programming skills, problem-solving abilities, and understanding of cybersecurity principles while providing hands-on experience in developing a real-world application.

The internship served as a valuable learning experience and provided a strong foundation for future software development and cybersecurity projects.

GitHub Repository

Repository Link:

https://github.com/Priya-6124/CyberSecurity_Internship

References

1. Python Official Documentation
2. Cyber Security Fundamentals
3. White Band Associates Internship Tasks
4. GitHub Repository Documentation